# Electric Field and Potential: Two Views of the Same Charge

Electric Field & Potential · oPhysics

| Senior Secondary (Years 11-12) | ~30 minutes | Science · Physics |
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Simulation: https://ophysics.com/em4.html
Curriculum links: electric fields

## Learning intentions

- I can distinguish an electric field (a vector quantity) from an electric potential (a scalar quantity).
- I can describe the relationship between field lines and equipotential lines.

## Getting started

Open the simulation with a single positive point charge. Show both the field lines and the equipotential lines using the checkboxes provided.

## Tasks

1. For a single positive charge, sketch the field lines and the equipotential lines. Describe the angle at which they cross.

2. Change to two like charges (both positive) of equal magnitude. Describe the shape of the field between them.

3. Change to two opposite charges (a dipole). Describe how the field and equipotential pattern differs from the two like charges in Task 2.

4. Using two charges of different magnitude, find a point where the electric field is exactly zero. Is this point closer to the smaller charge or the larger charge? Explain why.

## Extension (optional)

Explain, using what you observed, why two equipotential lines can never cross each other.